High performance Analog and Mixed signal VLSI System design

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Mahendra Sakare and Siva Rama Krishna

॥ सा विद्या या विमुक्तये ॥
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Indian Institute of Technology Dharwad

भारतीय प्रौद्योगिकी संस्थान रोपड़
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Course overview

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- Review of MOSFETs operation (Dr. Naveen K)
- Differential amplifiers and 2 stage opamp (Dr. Vinayak Hande)
- Compensation (stabilization) techniques (Dr. Vinayak Hande)
- Fully differential amplifiers (Dr. Naveen K)
- Voltage reference circuits (Dr. Vinayak Hande)
- Sample and hold circuits (Dr. Naveen K)
- Switched capacitor circuits (Dr. Naveen K)
- High speed comparators (Dr. Naveen K)
- Power supply design for mixed signal circuits (Dr. Vinayak Hande)
- High speed Clock data recovery circuits (Dr. Mahendra Sakare)
- CML circuit design (Dr. Mahendra Sakare)
- Layout of high speed circuits (Dr. Mahendra Sakare)
- Testing of custom integrated circuits (Dr. Mahendra Sakare)
- Wide tuning range CMOS VCOs (Industry talk: Dr. Siva Rama Krishna)
- Session on Values and ethics (Dr. Pandurangi)

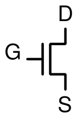
Introduction of experts

1. Vinayak Hande - Senior Researcher in RFEE Lab in Carinthia University of Applied Science
2. Mahendra Sakare - Assistant Prof. at IIT Ropar
3. Siva Rama Krishna - Staff Engineer at Mediatek Wireless Limited
4. Naveen Kadayinti - Assistant Prof. at IIT Dharwad
5. Dr. Aditya Pandurangi - Consulting Psychiatrist

Review of MOS operation

MOSFET operation in different bias conditions

- In an NMOS the source is usually grounded (i.e. the lowest potential)
- For $V_{GS} \leq V_{T_n}$, $\qquad I_{DS} = 0$
- For $V_{GS} > V_{T_n}$ the current depends V_{DS} and on the region of operation



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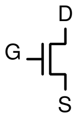
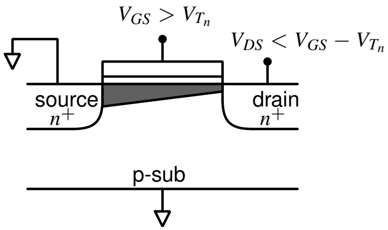


Linear region



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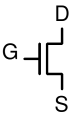

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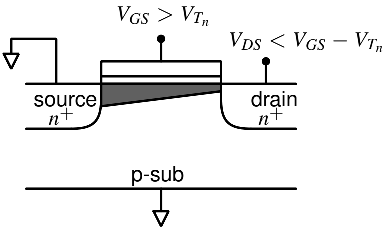

$$I_{DS} = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{T_n}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

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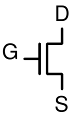
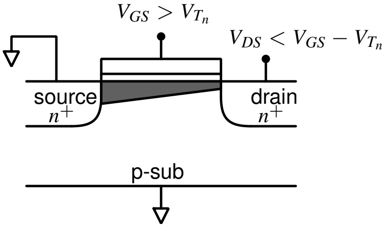


Linear region



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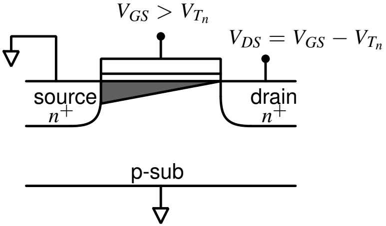
Edge of saturation region



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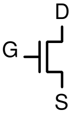
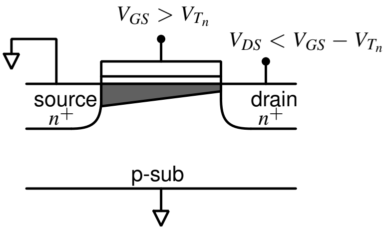


Linear region



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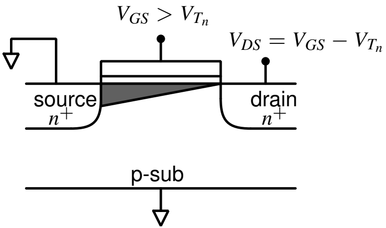
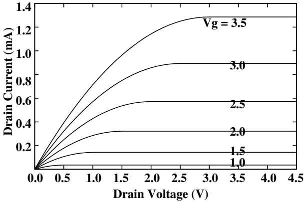
Edge of saturation region



$$I_{DS} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{T_n})^2$$

MOS model



- For $V_{GS} \leq V_{T_n}$,
  $I_{DS} = 0$
- For $V_{GS} > V_{T_n}$ and $V_{DS} \leq V_{GS} - V_{T_n}$,
  $I_{DS} = K\left[(V_{GS} - V_{T_n})V_{DS} - \frac{1}{2}V_{DS}^2\right]$
- For $V_{GS} > V_{T_n}$ and $V_{DS} > V_{GS} - V_{T_n}$,
  $I_{DS} = \frac{K}{2}(V_{GS} - V_{T_n})^2$

- In summary:
  $I_{DS} \propto K$, where, $K \equiv \mu_n C_{ox} \frac{W}{L}$ and $C_{ox} \equiv \frac{\epsilon_{ox}}{t_{ox}}$

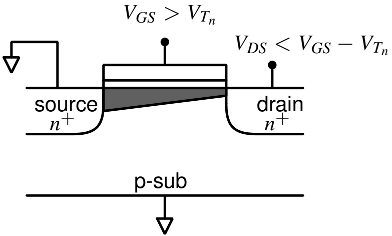
(Gate capacitance C_{ox} is *per unit area*)

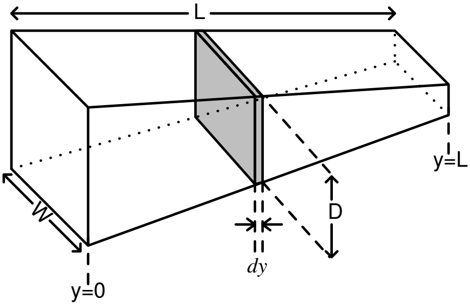
Gradual channel approximation

- Transistor's current voltage relations are very complex. A SPICE model of a transistor in 180 nm process has more than 100 model parameters. The number of model parameters is more than 250 in 65 nm CMOS and lower nodes.
- In order to analyze circuits, however, approximate models are useful. The approximate models have about 4-5 design parameters and can be analyzed easily and gives useful insights about circuits. The designs can then be verified with simulations.
- Gradual channel approximation is one such model of the MOSFET, in which we construct a 1-D model of the channel to solve for its current voltage characteristics.

GCA contd.

- We make the following assumptions
  - The threshold voltage is constant along the channel's width and length.
  - conduction of current is uniform along a section at any position.
- The channel resistance is not uniform along the length. Consider a small segment of the channel of width dy, as shown in the figure.





GCA contd.

In order to find an expression for the drain current, we need to find an expression for the incremental resistance of the channel region. The resistance of the channel is not constant and depends on the position along the y-axis.

GCA contd.

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The conductivity of the segment is given by

$$\sigma(y) = e\mu_n n(y)$$

where e is the charge of an electron, μ_n is the surface mobility of the channel. $n(y)$ is the density of charge carriers, which depends on the position y.

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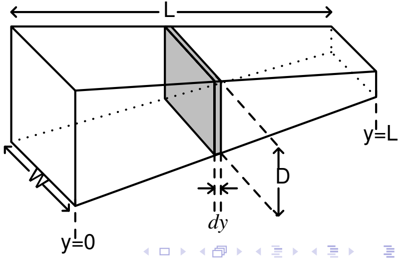
If $N(y)$ is the total number of carriers in the volume of the channel segment we can write,

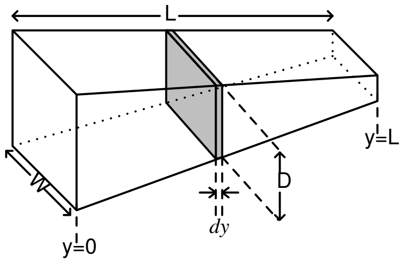
$$n(y) = \frac{N(y)}{WDdy}$$



GCA contd.

We can write

$$\sigma(y) = \frac{\mu_n eN(y)}{WDdy}$$



$eN(y)$ is the total charge in the channel segment $Q_T(y)$, which can be written as

$$Q_T(y) = Q(y)Wdy$$

where $Q(y) = C_{ox}[V_{GS} - V_{T_n} - V_c(y)]$ is the charge density per unit area of the channel.

GCA contd.

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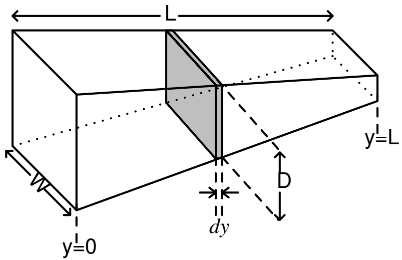
$$\sigma(y) = \frac{\mu_n e N(y)}{W D dy}$$



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Solving for $\sigma(y)$ we get,

$$\sigma(y) = \frac{\mu_n Q(y)}{D}$$

GCA contd.

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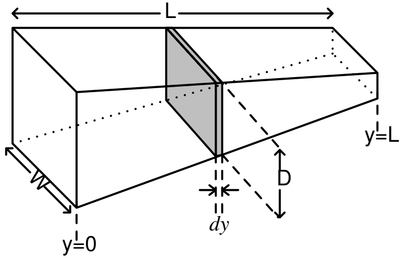
$$\sigma(y) = \frac{\mu_n eN(y)}{WDdy}$$



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Solving for $\sigma(y)$ we get,

$$\sigma(y) = \frac{\mu_n Q(y)}{D}$$

Now, we can write the resistance of the channel segment dR as

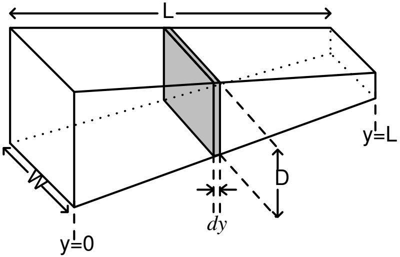
$$dR = \frac{1}{\sigma(y)}\frac{dy}{WD} = \frac{D}{\mu_n Q(y)}\frac{dy}{WD} = \frac{dy}{\mu_n Q(y)W}$$

GCA contd.

The drop across this channel segment can then be written as

$$dV_c = I_{DS}dR = I_{DS}\frac{dy}{\mu_n Q(y) W}$$

$$I_{DS}dy = \mu_n W Q(y) dV_c = \mu_n W C_{ox}(V_{GS} - V_{T_n} - V_c(y))dV_c$$



GCA contd.

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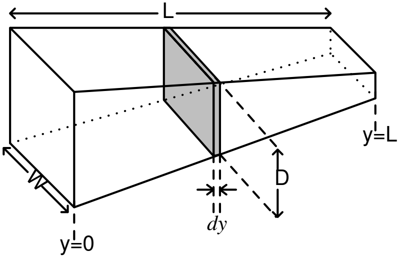

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$$I_{DS}dy = \mu_n W Q(y) dV_c = \mu_n W C_{ox}(V_{GS} - V_{T_n} - V_c(y))dV_c$$

Integrating this equation and using the boundary conditions, we get

$$\int_0^L I_{DS}dy = \mu_n C_{ox} W \int_0^{V_{DS}} (V_{GS} - V_{T_n} - V_c(y))dV_c$$

$$I_{DS} = \mu_n C_{ox}\frac{W}{L}\left[(V_{GS} - V_{T_n})V_{DS} - \frac{V_{DS}^2}{2}\right]$$

GCA contd.

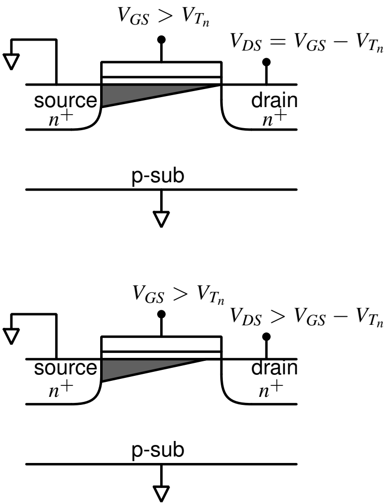
$$I_{DS} = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{T_n}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

As we increase the drain voltage, the channel at the drain end gets pinched off when $V_{DS} = V_{GS} - V_{T_n}$. This is the edge of saturation region. Under these conditions the drain current can then be written as

$$I_{DS} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{T_n})^2$$

If we increase the drain voltage further, the pinch-off point moves towards the source. This phenomenon is called Channel length modulation (or sometimes as Early effect - due to its similarity to Early effect in Bipolar transistors)

Channel length modulation aka Early effect



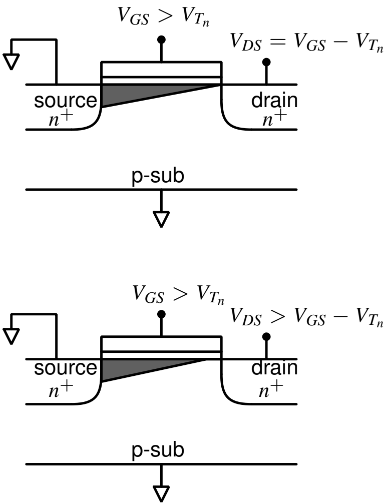
The inversion layer charge at the source end is

$$Q_I(y=0) = C_{ox}(V_{GS} - V_{T_n})$$

and at the drain end it is

$$Q_I(y=L) = C_{ox}(V_{GS} - V_{T_n} - V_{DS})$$

Channel length modulation aka Early effect



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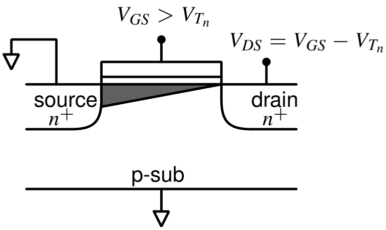
and at the drain end it is

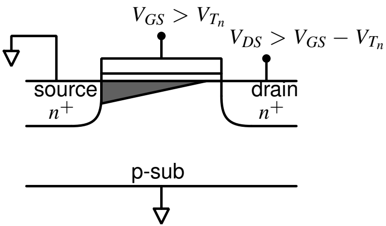
$$Q_I(y=L) = C_{ox}(V_{GS} - V_{T_n} - V_{DS})$$

At the edge of saturation, $V_{DS} = V_{GS} - V_{T_n}$ and

$$Q_I(y=L) = 0$$

Channel length modulation aka Early effect





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and at the drain end it is

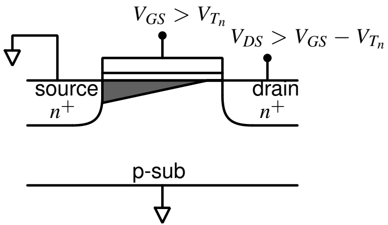
$$Q_I(y=L) = C_{ox}(V_{GS} - V_{T_n} - V_{DS})$$

At the edge of saturation, $V_{DS} = V_{GS} - V_{T_n}$ and

$$Q_I(y=L) = 0$$

If we make $V_{DS} > V_{GS} - V_{T_n}$, effective channel length reduces ($L' = L - \Delta L$)

Channel length modulation contd.



At $y = L'$, $Q_I(y) = 0$ and $V_c(y = L') = V_{Dsat}$
We can write

$$I_{DSat} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L'}(V_{GS} - V_{T_n})^2$$

Note: As V_{DS} increases, L' decreases, thus increasing the drain current.

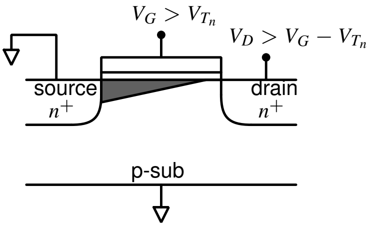
We can write

$$I_{DSat} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L - \Delta L}(V_{GS} - V_{T_n})^2 = \frac{1}{2}\mu_n C_{ox}\frac{W}{L(1 - \frac{\Delta L}{L})}(V_{GS} - V_{T_n})^2$$

If $\frac{\Delta L}{L}$ is very small, we can write this as

$$I_{DSat} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{T_n})^2\left(1 + \frac{\Delta L}{L}\right)$$

Channel length modulation contd.



Emperically the relation between ΔL and V_{DS} is given as

$$\frac{\Delta L}{L} \approx \lambda V_{DS}$$

where, λ is called the channel length modulation parameter.

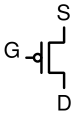
The equation for the drain current can then be written as

$$I_{DS} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{T_n})^2(1 + \lambda V_{DS})$$

Typically λ takes values around 0.01 for 180 nm technology nodes older than 180 nm. Scaled technologies have larger values of around 0.1 or more.

Current equations for PMOS transistors

- In a PMOS the source is generally tied to the highest potential.



- The current equations for PMOS transistors are
- For $V_{GS} \leq V_{T_p}$, (note that V_{T_p} is negative)
  $I_{DS} = 0$
- For $V_{GS} > V_{T_p}$ and $V_{DS} \leq V_{GS} - V_{T_p}$,
  $I_{DS} = K\left[(V_{GS} - V_{T_p})V_{DS} - \frac{1}{2}V_{DS}^2\right]$
- For $V_{GS} > V_{T_p}$ and $V_{DS} > V_{GS} - V_T$,
  $I_{DS} = \frac{K}{2}(V_{GS} - V_{T_p})^2$

Current equations for PMOS transistors contd.

- If you find it difficult to grapple with the signs of quantities in PMOS transistors equations, you can simply use modulus of all values and note that current flows from higher potential to lower potential.
- i.e. the current equations can be written as follows
- For $|V_{GS}| \leq |V_{T_p}|$,
  $|I_{DS}| = 0$
- For $|V_{GS}| > |V_{T_p}|$ and $|V_{DS}| \leq (|V_{GS}| - |V_{T_p}|)$,
  $|I_{DS}| = K\left[(|V_{GS}| - |V_{T_p}|)|V_{DS}| - \frac{1}{2}|V_{DS}|^2\right]$
- For $|V_{GS}| > |V_{T_p}|$ and $|V_{DS}| > |V_{GS}| - |V_{T_p}|$,
  $|I_{DS}| = \frac{K}{2}(|V_{GS}| - |V_{T_p}|)^2$

Transconductance of the MOSFET

- Transconductance and output impedance are two important parameters of transistors for most analog circuits.
- Transconductance is basically change in drain current for a small change in gate-source voltage. This can be computed as

$$\begin{aligned} I_{DS} &= \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{T_n})^2, \\ g_m &= \frac{\partial I_{DS}}{\partial V_{GS}} \\ &= \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{T_n}). \end{aligned}$$

- We can subsctitute for $(V_{GS} - V_{T_n})$ from the current equation and write g_m as

$$g_m = \sqrt{2\mu_n C_{ox}\frac{W}{L}I_{DS}}.$$

Transconductance of the MOSFET

- $$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{T_n}).$$
  $$g_m = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_{DS}}.$$
- We can also substitute for $\mu_n C_{ox} \frac{W}{L}$ from the current equation and exress g_m as
  $$g_m = \frac{2 I_{DS}}{V_{GS} - V_{T_n}}.$$
- These equations contradict each other! Which equation should we use for calculating the transconductance?

Output impedance

- The small signal output conductance of the MOSFET can be calculated from the current equation as

$$\begin{aligned} g_o &= \frac{\partial I_{DS}}{\partial V_{DS}} \\ &= \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{T_n})^2(\lambda V_{DS}) \\ &= \lambda I_{DS}. \end{aligned}$$

- The output impedance is therefore

$$r_o = \frac{1}{\lambda I_{DS}}.$$

References

1. Textbook: Sung-Mo Kang, Yusuf Leblebici, “CMOS Digital Integrated Circuits Analysis & Design”
2. Textbook: R. Jacob Baker, “CMOS circuit design, layout and simulation”